

Lecture 14: Treatment effects and difference-in-differences

Economics 326 — Introduction to Econometrics II

Vadim Marmer, UBC

Motivation: causal questions

- Many important questions in economics are **causal**:
 - Does job training increase earnings?
 - Does a new drug improve health outcomes?
 - Does building an incinerator reduce nearby house prices?
- The **fundamental challenge**: we can never observe the same individual both with and without treatment at the same time.

Potential outcomes

- For each individual i , define two **potential outcomes**:
 - $Y_i(1)$: outcome if individual i receives treatment ($D_i = 1$),
 - $Y_i(0)$: outcome if individual i does not receive treatment ($D_i = 0$).
- The **individual treatment effect** for person i is:

$$Y_i(1) - Y_i(0).$$

- Example: if Y_i is earnings and D_i indicates job training, then $Y_i(1) - Y_i(0)$ is the causal effect of training on earnings for person i .

The fundamental problem

- Let $D_i \in \{0, 1\}$ denote treatment status. The **observed outcome** is:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

- If $D_i = 1$, we observe $Y_i(1)$ but not $Y_i(0)$.
- If $D_i = 0$, we observe $Y_i(0)$ but not $Y_i(1)$.
- We can never observe both potential outcomes for the same individual. This is the **fundamental problem of causal inference**.

Average treatment effects

- Since individual treatment effects $Y_i(1) - Y_i(0)$ are unobservable, we focus on averages.
- **Average Treatment Effect (ATE)**:

$$\text{ATE} = E[Y_i(1) - Y_i(0)].$$

- **Average Treatment Effect on the Treated (ATT):**

$$\text{ATT} = \text{E}[Y_i(1) - Y_i(0) \mid D_i = 1].$$

- ATE averages over the entire population; ATT averages only over those who actually receive treatment.

Selection bias

- A naive comparison of treated and untreated outcomes yields:

$$\begin{aligned} & \text{E}[Y_i \mid D_i = 1] - \text{E}[Y_i \mid D_i = 0] \\ &= \text{E}[Y_i(1) \mid D_i = 1] - \text{E}[Y_i(0) \mid D_i = 0]. \end{aligned}$$

- Add and subtract $\text{E}[Y_i(0) \mid D_i = 1]$:

$$\begin{aligned} &= \underbrace{\text{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} \\ &\quad + \underbrace{\text{E}[Y_i(0) \mid D_i = 1] - \text{E}[Y_i(0) \mid D_i = 0]}_{\text{Selection bias}}. \end{aligned}$$

- **Selection bias** arises when the treatment and control groups differ in their baseline outcomes $Y_i(0)$. For example, people who choose to enroll in job training may differ systematically from those who do not.

Random assignment

- Under **random assignment**, treatment D_i is independent of potential outcomes:

$$\text{E}[Y_i(0) \mid D_i = 1] = \text{E}[Y_i(0) \mid D_i = 0] = \text{E}[Y_i(0)].$$

- The selection bias vanishes, and the simple difference in means equals the ATE:

$$\text{E}[Y_i \mid D_i = 1] - \text{E}[Y_i \mid D_i = 0] = \text{ATE} = \text{ATT}.$$

- Randomized experiments (like the National Supported Work program) achieve this by randomly assigning individuals to treatment and control groups.

Regression with a treatment dummy

- With random assignment, the ATE can be estimated by regressing Y_i on a treatment dummy:

$$Y_i = \alpha + \tau D_i + U_i,$$

where $\text{E}[U_i \mid D_i] = 0$ under randomization.

- The OLS estimate of τ equals the difference in sample means:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0,$$

where $\bar{Y}_1$ and $\bar{Y}_0$ are the sample averages for the treated and control groups.

Example: Lalonde data

- The **National Supported Work (NSW)** program randomly assigned disadvantaged workers to receive job training.
- We use the `jtrain2` dataset from the `wooldridge` package, based on the Lalonde (1986) study:

```
library(wooldridge)
data(jtrain2)
head(jtrain2[, c("train", "re78", "educ", "age", "black", "married")], n = 10)
```

	train	re78	educ	age	black	married
1	1	9.93005	11	37	1	1
2	1	3.59589	9	22	0	0
3	1	24.90950	12	30	1	0
4	1	7.50615	11	27	1	0
5	1	0.28979	8	33	1	0
6	1	4.05649	9	22	1	0
7	1	0.00000	12	23	1	0
8	1	8.47216	11	32	1	0
9	1	2.16402	16	22	1	0
10	1	12.41810	12	33	0	1

- `train`: 1 if assigned to job training, 0 otherwise.
- `re78`: real earnings in 1978 (in dollars).

Estimating the ATE

- Since treatment was randomly assigned, the coefficient on `train` estimates the ATE:

```
summary(lm(re78 ~ train, data = jtrain2))$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.554802	0.4080460	11.162474	1.154113e-25
train	1.794343	0.6328536	2.835321	4.787524e-03

- The estimated ATE is approximately \$1,794: on average, participation in the job training program increased 1978 earnings by about \$1,794.

Observational studies

- In many settings, treatment is **not randomly assigned**. Individuals self-select into treatment.
- Example: workers who voluntarily enroll in job training may be more motivated or have lower baseline earnings than those who do not enroll.
- Self-selection generates **selection bias**, and the simple difference in means no longer estimates the ATE.
- To estimate treatment effects from observational data, we need to **control for covariates** that affect both treatment selection and outcomes.

Potential outcomes with a covariate

- Suppose the potential outcomes depend linearly on a covariate X_i :

$$Y_i(0) = \alpha_0 + \beta_0 X_i + U_i(0),$$

$$Y_i(1) = \alpha_1 + \beta_1 X_i + U_i(1),$$

where $E[U_i(0) \mid X_i, D_i] = 0$ and $E[U_i(1) \mid X_i, D_i] = 0$.

- These assumptions mean that, after conditioning on X_i , treatment assignment D_i is as good as random. This is called **conditional mean independence** (or “selection on observables”).

The ATE with a covariate

- Taking expectations of the potential outcomes:

$$\begin{aligned} E[Y_i(1)] &= \alpha_1 + \beta_1 E[X_i], \\ E[Y_i(0)] &= \alpha_0 + \beta_0 E[X_i]. \end{aligned}$$

- The ATE is:

$$\begin{aligned} \text{ATE} &= E[Y_i(1) - Y_i(0)] \\ &= (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)E[X_i]. \end{aligned}$$

- The ATE depends on the difference in intercepts **and** the difference in slopes, weighted by the population mean of X_i .

Two separate regressions

- One approach: estimate separate regressions for each group.
- **Control group** ($D_i = 0$): $Y_i = \alpha_0 + \beta_0 X_i + U_i(0)$.
- **Treatment group** ($D_i = 1$): $Y_i = \alpha_1 + \beta_1 X_i + U_i(1)$.
- The estimated ATE is:

$$\begin{aligned} \widehat{\text{ATE}} &= (\hat{\alpha}_1 - \hat{\alpha}_0) \\ &\quad + (\hat{\beta}_1 - \hat{\beta}_0)\bar{X}, \end{aligned}$$

where $\bar{X}$ is the overall sample mean of X_i .

Combined regression with interactions

- The observed outcome $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$ can be written as a single regression. Expanding:

$$\begin{aligned} Y_i &= \alpha_0(1 - D_i) + \alpha_1 D_i \\ &\quad + \beta_0 X_i(1 - D_i) + \beta_1 X_i D_i + \tilde{U}_i \\ &= \alpha_0 + (\alpha_1 - \alpha_0) D_i \\ &\quad + \beta_0 X_i + (\beta_1 - \beta_0) X_i D_i + \tilde{U}_i, \end{aligned}$$

where $\tilde{U}_i = (1 - D_i)U_i(0) + D_i U_i(1)$.

- This is a regression of Y_i on D_i , X_i , and the interaction $X_i D_i$.
- The coefficient on D_i is $\alpha_1 - \alpha_0$, which is **not** the ATE (unless $\beta_1 = \beta_0$).

The demeaning trick

- Write $X_i = E[X_i] + (X_i - E[X_i])$. Then the interaction term becomes:

$$\begin{aligned}(\beta_1 - \beta_0)X_i D_i &= (\beta_1 - \beta_0)E[X_i] D_i \\ &\quad + (\beta_1 - \beta_0)(X_i - E[X_i])D_i.\end{aligned}$$

- Substituting into the regression:

$$\begin{aligned}Y_i &= \alpha_0 + \left[(\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)E[X_i] \right] D_i \\ &\quad + \beta_0 X_i + (\beta_1 - \beta_0)(X_i - E[X_i])D_i + \tilde{U}_i.\end{aligned}$$

- Define $\tau = (\alpha_1 - \alpha_0) + (\beta_1 - \beta_0)E[X_i]$ and $\delta = \beta_1 - \beta_0$:

$$\begin{aligned}Y_i &= \alpha_0 + \tau D_i + \beta_0 X_i \\ &\quad + \delta(X_i - E[X_i])D_i + \tilde{U}_i.\end{aligned}$$

- The coefficient τ on D_i is exactly the **ATE**.

Estimating the ATE with covariates

- In practice, replace $E[X_i]$ with the sample mean $\bar{X}$ and run the regression:

$$\begin{aligned}Y_i &= \hat{\alpha}_0 + \hat{\tau} D_i + \hat{\beta}_0 X_i \\ &\quad + \hat{\delta}(X_i - \bar{X})D_i + \text{residual}.\end{aligned}$$

- The coefficient $\hat{\tau}$ on D_i directly estimates the ATE.
- This works because demeaning the interaction “absorbs” the $(\beta_1 - \beta_0)E[X_i]$ part of the ATE into the coefficient on D_i .

Why not regress Y_i on D_i and X_i ?

- A simpler regression omits the interaction:

$$Y_i = a + \tau D_i + b X_i + V_i.$$

- This is valid if $\beta_1 = \beta_0$ (the covariate has the same effect in both groups). Then $\delta = 0$, the interaction drops out, and the coefficient on D_i is the ATE.
- If $\beta_1 \neq \beta_0$, the omitted interaction creates bias. The regression with the demeaned interaction nests the simpler model as a special case.

Example: separate regressions

- Estimate separate regressions of `re78` on `educ` for the treatment and control groups:

```
reg0 <- lm(re78 ~ educ, data = jtrain2, subset = (train == 0))
reg1 <- lm(re78 ~ educ, data = jtrain2, subset = (train == 1))
cbind(Control = coef(reg0), Treatment = coef(reg1))
```

	Control	Treatment
(Intercept)	3.80301658	-0.7821703
educ	0.07451936	0.6892860

- Compute the estimated ATE:

```
xbar <- mean(jtrain2$educ)
a0 <- coef(reg0)[1]; b0 <- coef(reg0)[2]
a1 <- coef(reg1)[1]; b1 <- coef(reg1)[2]
ATE_manual <- (a1 - a0) + (b1 - b0) * xbar
cat("Sample mean of educ:", round(xbar, 2), "\n")
```

Sample mean of educ: 10.2

```
cat("Estimated ATE:", round(ATE_manual, 2), "\n")
```

Estimated ATE: 1.68

Example: demeaned regression

- Create the demeaned interaction and run the combined regression:

```
jtrain2$educ_dm <- jtrain2$educ - mean(jtrain2$educ)
reg_dm <- lm(re78 ~ train + educ + I(train * educ_dm), data = jtrain2)
summary(reg_dm)$coefficients
```

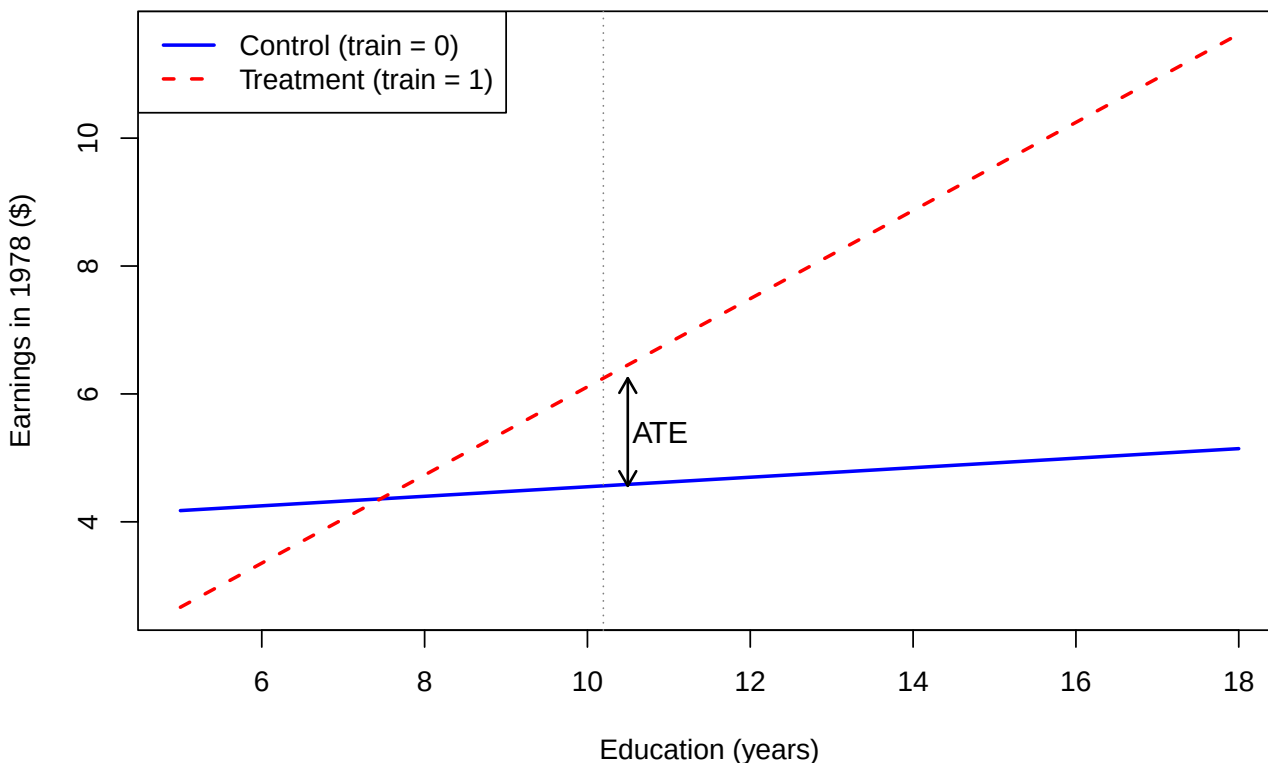
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.80301658	2.5689136	1.4803988	0.13948090
train	1.68266981	0.6299604	2.6710725	0.00784062
educ	0.07451936	0.2514521	0.2963561	0.76709766
I(train * educ_dm)	0.61476663	0.3472755	1.7702560	0.07737534

- The coefficient on `train` directly estimates the ATE, matching the result from the separate regressions approach.

Example: regression lines

- The two regression lines, with a vertical line at $\bar{X}$ and the ATE marked as the gap:

Separate regression lines by treatment group



Example: adding more covariates

- So far, we used only `educ` as the control. The dataset contains more pre-treatment characteristics: `age`, `black`, `married`.
- With random assignment, slopes are approximately equal across groups, so we can add controls directly without demeaned interactions:

```
reg_X <- lm(re78 ~ train + educ + age + black + married, data = jtrain2)
round(summary(reg_X)$coefficients, 1)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9	2.2	0.4	0.7
train	1.7	0.6	2.7	0.0
educ	0.4	0.2	2.4	0.0
age	0.1	0.0	1.2	0.2
black	-2.3	0.8	-2.7	0.0
married	0.2	0.8	0.2	0.9

Comparison: simple vs. controlled

- Compare the estimated ATE and its standard error across specifications:

```
m1 <- lm(re78 ~ train, data = jtrain2)
m2 <- lm(re78 ~ train + educ + age + black + married, data = jtrain2)
tab <- rbind(
  "No controls" = c(Estimate = coef(m1)["train"],
                    SE       = summary(m1)$coef["train", "Std. Error"]),
  "With controls" = c(Estimate = coef(m2)["train"],
                      SE       = summary(m2)$coef["train", "Std. Error"])
)
round(tab, 0)
```

	Estimate	train	SE
No controls	2	1	
With controls	2	1	

- The estimated ATE barely changes. The standard error shrinks when controls are added.

Why do controls change little here?

- Under **random assignment**, D_i is independent of all covariates: the treatment and control groups are balanced in expectation.
- Because of balance, including X_i does not change the coefficient on D_i : there is no selection bias to remove.

Why does the standard error shrink?

- Recall that the variance of the OLS estimator depends on the residual variance $\hat{\sigma}^2$:

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{\hat{\sigma}^2}{\sum_{i=1}^n (D_i - \bar{D})^2 (1 - R_D^2)},$$

where R_D^2 is the R^2 from regressing D_i on the other regressors.

- With randomization, D_i is nearly uncorrelated with covariates, so $R_D^2 \approx 0$: the denominator stays roughly the same.
- However, covariates that predict earnings absorb variation in Y_i , reducing $\hat{\sigma}^2$. Since $\text{se}(\hat{\tau}) \propto \hat{\sigma}$, the standard error shrinks.
- In short: **adding covariates under randomization buys precision without changing the point estimate.**

Example: observational data (jtrain3)

- The `jtrain3` dataset from the `wooldridge` package combines the same NSW trainees with a **non-experimental** control group drawn from the Current Population Survey (CPS).
- In this dataset, treatment is **not** randomly assigned: CPS respondents differ systematically from NSW participants.

```
data(jtrain3)
cat("n =", nrow(jtrain3), " (train=1:", sum(jtrain3$train == 1),
    ", train=0:", sum(jtrain3$train == 0), ")\n")
```

```
n = 2675 (train=1: 185 , train=0: 2490 )
```

Observational: without controls

- A simple regression of earnings on the training dummy:

```
obs1 <- lm(re78 ~ train, data = jtrain3)
round(summary(obs1)$coefficients, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	21.554	0.304	70.985	0
train	-15.205	1.155	-13.169	0

- The estimated “effect” is **large and negative**: trainees appear to earn far less than the CPS comparison group.
- This is pure **selection bias**: individuals selected into the NSW program were disadvantaged, with much lower baseline earnings.

Observational: with controls

- Adding covariates dramatically changes the estimate:

```
obs2 <- lm(re78 ~ train + educ + age + black + hisp + married + re74 + re75,
           data = jtrain3)
round(summary(obs2)$coefficients, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.777	1.366	0.569	0.570
train	0.860	0.908	0.947	0.344
educ	0.528	0.075	7.015	0.000
age	-0.082	0.021	-3.940	0.000
black	-0.543	0.494	-1.098	0.272
hisp	2.166	1.092	1.983	0.048
married	1.220	0.586	2.083	0.037
re74	0.278	0.028	9.952	0.000
re75	0.568	0.028	20.615	0.000

- The coefficient on `train` shifts from about \$-\$15,200 to about \$+\$900, much closer to the experimental benchmark of \$+\$1,794.

Experimental vs. observational

- Side-by-side comparison:

```
make_row <- function(m) c(Estimate = coef(m)["train"],
                          SE = summary(m)$coef["train", "Std. Error"])
tab <- rbind(
  "Experimental: no controls" = make_row(m1),
  "Experimental: with controls" = make_row(m2),
  "Observational: no controls" = make_row(obs1),
  "Observational: with controls" = make_row(obs2)
)
round(tab, 3)
```

	Estimate.train	SE
Experimental: no controls	1.794	0.633
Experimental: with controls	1.679	0.629
Observational: no controls	-15.205	1.155
Observational: with controls	0.860	0.908

- Experimental data:** controls barely change the estimate (1.79 vs. 1.68); standard errors shrink.
- Observational data:** controls remove most of the selection bias (\$-\$15.2 $\rightarrow$ \$+\$0.9); covariates are essential for a credible estimate.

From cross-sections to panel data

- So far, we used **cross-sectional** data and assumed selection on observables: after controlling for X_i , treatment is as good as random.
- In some settings, this assumption is hard to justify. An alternative approach exploits **panel data** (repeated observations on the same units over time).
- The **difference-in-differences (DID)** method compares changes over time between a treatment group and a control group.

DID setup

- Two time periods: $t \in \{0, 1\}$ (before and after treatment).

- Two groups: $D_i \in \{0, 1\}$ (control and treatment).
- Treatment occurs between periods 0 and 1, and only the treatment group ($D_i = 1$) is affected.
- We observe Y_{it} : the outcome for individual i at time t .

DID regression model

- The DID regression is:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta(t \cdot D_i) + U_{it},$$

where $E[U_{it} \mid D_i] = 0$.

- The 2×2 table of conditional means:

	$D_i = 0$ (Control)	$D_i = 1$ (Treatment)
$t = 0$	α	$\alpha + \gamma$
$t = 1$	$\alpha + \delta$	$\alpha + \delta + \gamma + \beta$

Interpreting the coefficients

- α : baseline expected outcome (control group, before treatment).
- δ : **time effect** — the change in the control group's outcome from $t = 0$ to $t = 1$. This captures common trends (e.g., inflation, economic growth).
- γ : **group difference** at baseline — the pre-existing difference between treatment and control groups at $t = 0$.
- β : **DID estimand** — the additional change in the treatment group's outcome, beyond what the control group experienced.

Deriving the DID estimand

- For each combination of t and D_i , take expectations using $E[U_{it} \mid D_i] = 0$:

$$\begin{aligned} t = 0, D_i = 0: & \quad E[Y_{i0} \mid D_i = 0] = \alpha, \\ t = 1, D_i = 0: & \quad E[Y_{i1} \mid D_i = 0] = \alpha + \delta, \\ t = 0, D_i = 1: & \quad E[Y_{i0} \mid D_i = 1] = \alpha + \gamma, \\ t = 1, D_i = 1: & \quad E[Y_{i1} \mid D_i = 1] = \alpha + \delta + \gamma + \beta. \end{aligned}$$

- Subtracting the control group change from the treatment group change:

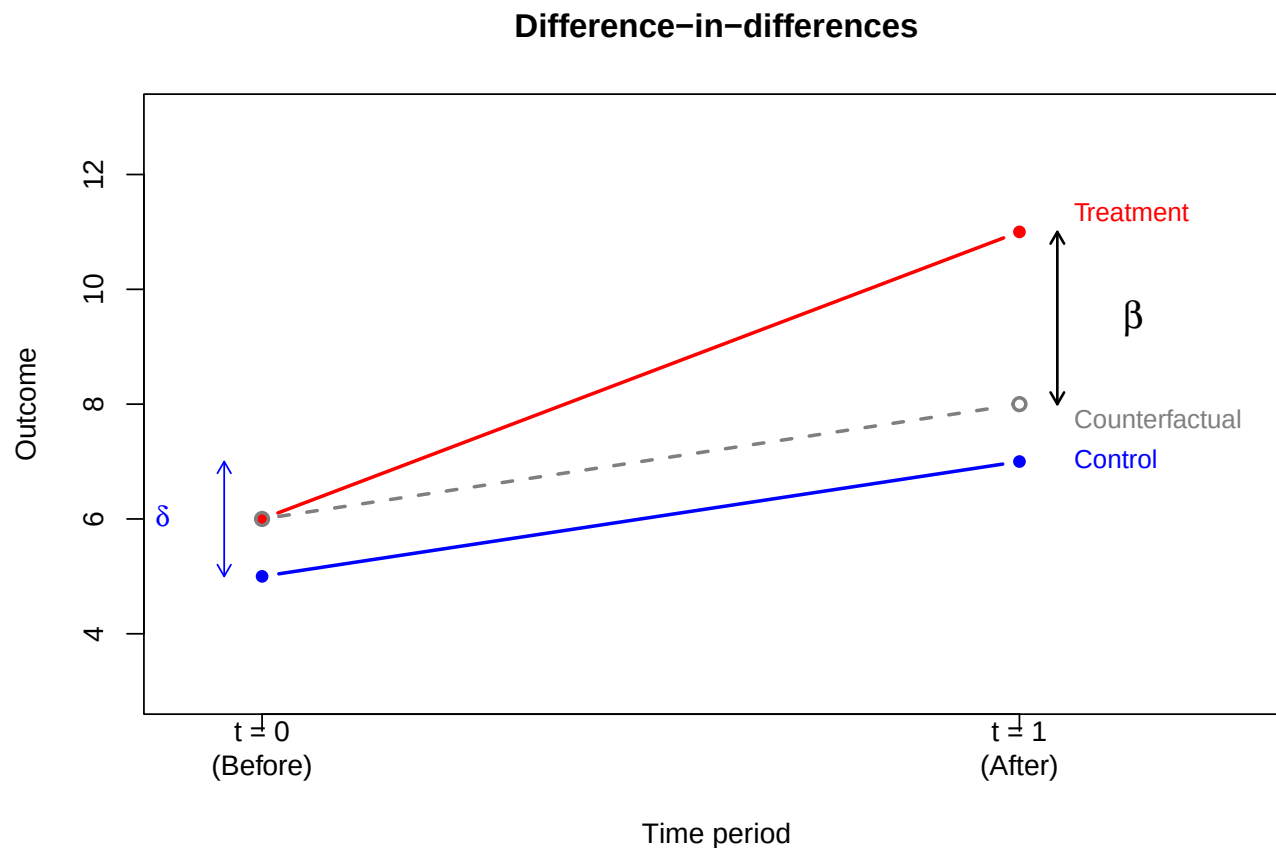
$$\begin{aligned} \beta = & \left(E[Y_{i1} \mid D_i = 1] - E[Y_{i0} \mid D_i = 1] \right) \\ & - \left(E[Y_{i1} \mid D_i = 0] - E[Y_{i0} \mid D_i = 0] \right). \end{aligned}$$

- The DID estimand is the **difference in within-group changes** over time:

$$\begin{aligned} \beta = & E[Y_{i1} - Y_{i0} \mid D_i = 1] \\ & - E[Y_{i1} - Y_{i0} \mid D_i = 0]. \end{aligned}$$

DID diagram

- The classic DID diagram shows the control group, treatment group, and counterfactual:



Example: incinerator and house prices

- Kiel and McClain (1995) studied how the construction of a garbage incinerator affected nearby house prices in North Andover, Massachusetts.
- We use the `kielmc` dataset from the `wooldridge` package:

```
data(kielmc)
head(kielmc[, c("rprice", "y81", "nearinc", "y81nrinc", "age")], n = 10)
```

	rprice	y81	nearinc	y81nrinc	age
1	60000	0	1	0	48
2	40000	0	1	0	83
3	34000	0	1	0	58
4	63900	0	1	0	11
5	44000	0	1	0	48
6	46000	0	1	0	78
7	56000	0	1	0	22
8	38500	0	1	0	78
9	60500	0	1	0	42
10	55000	0	1	0	41

- `rprice`: house price in 1978 dollars.
- `y81`: 1 if year is 1981 (after incinerator announced), 0 if 1978.
- `nearinc`: 1 if house is near the incinerator site.
- `y81nrinc`: interaction `y81 * nearinc`.

The 2×2 table of means

- Compute the four group means:

```
means <- tapply(kielmc$rprice, list(kielmc$y81, kielmc$nearinc), mean)
colnames(means) <- c("Far (nearinc=0)", "Near (nearinc=1)")
rownames(means) <- c("1978 (y81=0)", "1981 (y81=1)")
round(means)
```

	Far (nearinc=0)	Near (nearinc=1)
1978 (y81=0)	82517	63693
1981 (y81=1)	101308	70619

- Computing the DID by hand:

```
diff_near <- means[2, 2] - means[1, 2]
diff_far <- means[2, 1] - means[1, 1]
DID <- diff_near - diff_far
cat("Change (near):", round(diff_near), "\n")
```

Change (near): 6926

```
cat("Change (far): ", round(diff_far), "\n")
```

Change (far): 18790

```
cat("DID: ", round(DID), "\n")
```

DID: -11864

DID regression

- The DID regression:

```
reg_did <- lm(rprice ~ y81 + nearinc + y81nrinc, data = kielmc)
summary(reg_did)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	82517.23	2726.910	30.260341	1.709246e-95
y81	18790.29	4050.065	4.639502	5.116892e-06
nearinc	-18824.37	4875.322	-3.861154	1.368017e-04
y81nrinc	-11863.90	7456.646	-1.591051	1.125948e-01

- The coefficient on `y81nrinc` matches the DID computed from the 2×2 table.
- The estimated effect is negative (incinerator reduced nearby prices), but the p-value is around 0.11, so it is not statistically significant at the 5% level.

DID with covariates

- Adding house characteristics as controls can improve precision:

```
reg_did_cov <- lm(rprice ~ y81 + nearinc + y81nrinc + age + I(age^2),
                  data = kielmc)
summary(reg_did_cov)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	89116.535375	2406.0510717	37.038505	8.247920e-117
y81	21321.041753	3443.6310979	6.191442	1.857145e-09
nearinc	9397.935862	4812.2218389	1.952931	5.171307e-02
y81nrinc	-21920.269951	6359.7453905	-3.446721	6.444775e-04
age	-1494.424046	131.8603155	-11.333387	3.347719e-25
I(age^2)	8.691277	0.8481268	10.247615	1.859361e-21

- After controlling for house age, the DID estimate becomes larger in magnitude and statistically significant.
- Controlling for covariates reduces residual variance, leading to more precise estimates.

DID and potential outcomes

- To connect DID with the potential outcomes framework, define **panel potential outcomes**: $Y_{it}(d)$ is the outcome for individual i at time t if assigned to group $d \in \{0, 1\}$.
- The observed outcome is:

$$Y_{it} = D_i Y_{it}(1) + (1 - D_i) Y_{it}(0).$$

- What we observe for each group:

	Control ($D_i = 0$)	Treatment ($D_i = 1$)
$t = 0$	$Y_{i0}(0)$	$Y_{i0}(1)$
$t = 1$	$Y_{i1}(0)$	$Y_{i1}(1)$

- The treatment effect at time $t = 1$ for the treated group is:

$$ATT = E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1].$$

The counterfactual $Y_{i1}(0)$ is unobserved for the treated group.

DID as a treatment effect

- Recall that $\beta = E[Y_{i1} - Y_{i0} \mid D_i = 1] - E[Y_{i1} - Y_{i0} \mid D_i = 0]$.
- Substituting observed outcomes with potential outcomes:

$$\begin{aligned} \beta &= E[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1] \\ &\quad - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]. \end{aligned}$$

- To relate β to the ATT, add and subtract $E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1]$:

$$\begin{aligned} \beta &= \underbrace{E[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]}_{ATT} \\ &\quad + E[Y_{i1}(0) - Y_{i0}(1) \mid D_i = 1] \\ &\quad - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]. \end{aligned}$$

- For β to equal the ATT, we need two additional assumptions.

Assumption 1: no anticipation

- **No anticipation:** the outcome at $t = 0$ (before treatment) is not affected by future treatment group assignment:

$$E[Y_{i0}(1) \mid D_i = 1] = E[Y_{i0}(0) \mid D_i = 1].$$

- Being assigned to the treatment group does not change pre-treatment outcomes in expectation.
- In the incinerator example: before the incinerator was announced, living near the future site did not affect house prices (relative to what they would have been otherwise).
- Under no anticipation, we can replace $Y_{i0}(1)$ with $Y_{i0}(0)$ in the expression for β :

$$\beta = \text{ATT} + E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] - E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0].$$

Assumption 2: parallel trends

- **Parallel trends:** in the absence of treatment, both groups would have experienced the same change over time:

$$E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0] = E[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1].$$

- Under both no anticipation and parallel trends:

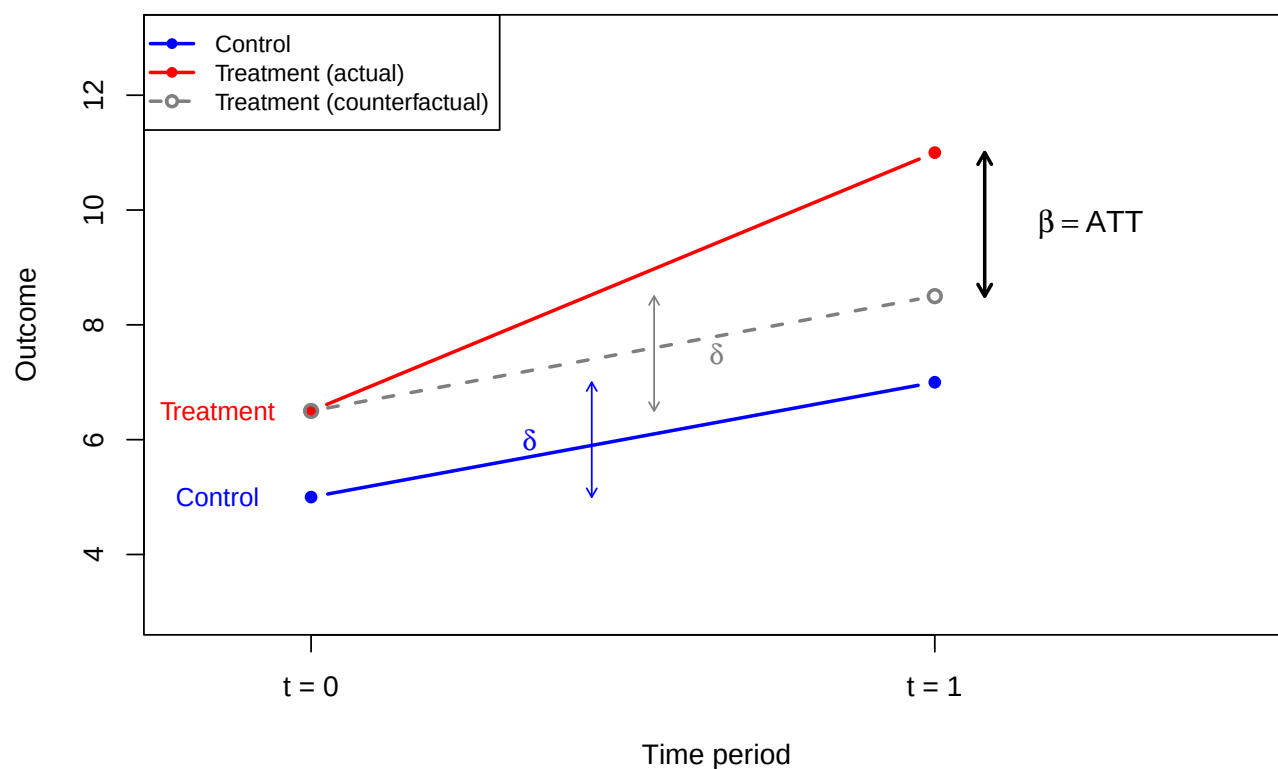
$$\beta = \text{ATT}.$$

- The parallel trends assumption cannot be directly tested because $Y_{i1}(0)$ is unobserved for the treated group. However, if pre-treatment data for multiple periods exist, one can check whether trends were parallel before treatment.

Parallel trends diagram

- Illustrating the parallel trends assumption:

Parallel trends assumption



Summary

- **Potential outcomes** $Y_i(1)$ and $Y_i(0)$ formalize causal effects. The individual treatment effect $Y_i(1) - Y_i(0)$ is never fully observed.
- The **ATE** and **ATT** are population-level summaries of treatment effects.
- With **random assignment**, a simple regression of Y_i on D_i estimates the ATE.
- With **observational data**, controlling for covariates and using the **demeaning trick** (interacting D_i with $X_i - \bar{X}$) allows the coefficient on D_i to estimate the ATE.
- **Difference-in-differences** uses panel data to compare changes over time between groups:

$$\beta = \text{E}[Y_{i1} - Y_{i0} \mid D_i = 1] \\ - \text{E}[Y_{i1} - Y_{i0} \mid D_i = 0].$$

- Under **no anticipation** and **parallel trends**, the DID estimand β equals the ATT.